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COMP4358(1) - Cloud Computing
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NSG Configuration Lab Report

Limiting SSH Access by IP Address Using Azure Network Security Groups

Overview

This lab configured two Azure Virtual Machines in a jump-host architecture. VM1 is the public-facing entry point with its Network Security Group (NSG) allowing SSH only from one specific IP address. VM2 sits on a private subnet with no public IP and is only reachable via SSH from VM1's private IP. The objective was to implement IP-based SSH access control using Azure NSGs and verify it end-to-end.

Issues Encountered

1. SSH to VM1 Returning Permission Denied

After creating the nau_ip_limit NSG rule on VM1, SSH from the local machine returned "Permission denied, please try again" and "Connection reset by 20.51.217.204 port 22." The source IP in the NSG rule had become outdated — the rule was holding a previous public IP that no longer matched the machine's current address. The current IP was verified at whatismyipaddress.com and the rule was updated to 104.55.71.203, after which SSH connected successfully.

2. VM1 and VM2 on Different Virtual Networks

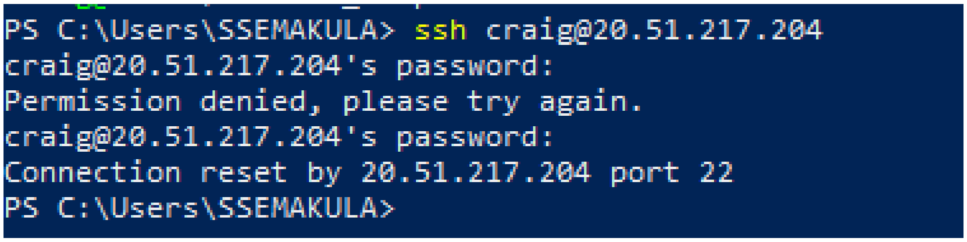
When VM2 was first created it was placed on vnet-eastus-2 while VM1 was on vnet-eastus-1. VMs on separate VNets cannot reach each other by private IP — there is no network path between them. All SSH attempts from VM1 to VM2's private IP timed out. VM2 was deleted and recreated inside vnet-eastus-1 with a new subnet snet-eastus-2 (172.17.1.0/24), placing both VMs on the same virtual network.

3. VM2 Had an Unintended Public IP

During initial VM2 creation, a public IP (172.174.155.67) was automatically assigned. The assignment requires VM2 to be accessible only through VM1 — having a public IP exposes it directly to the internet. The public IP was removed by setting the Public IP field to None when recreating VM2.

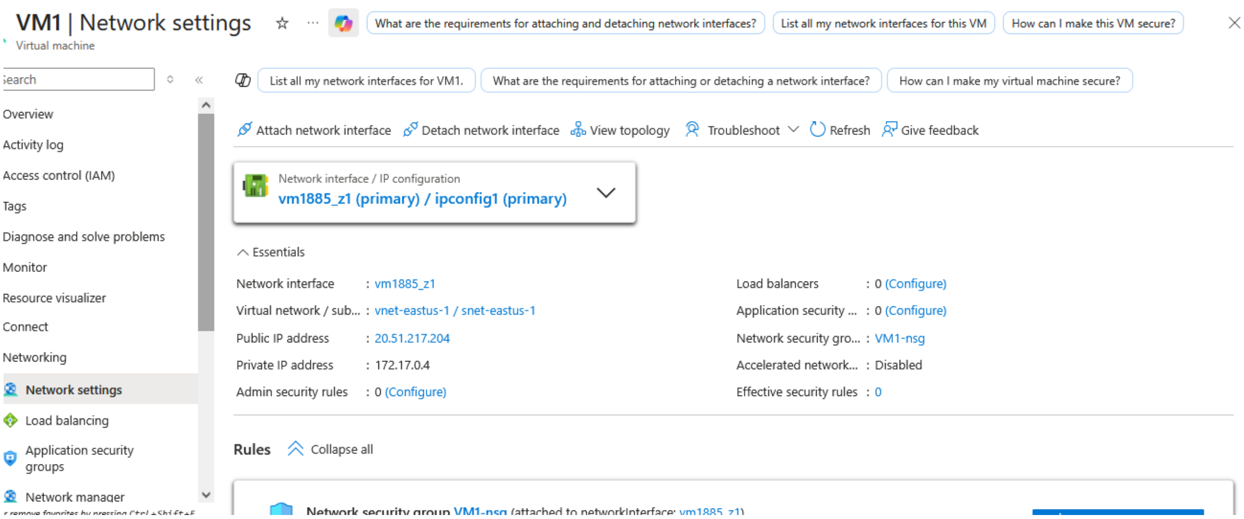
Screenshots

Fig 1: SSH to VM1 returning Permission Denied



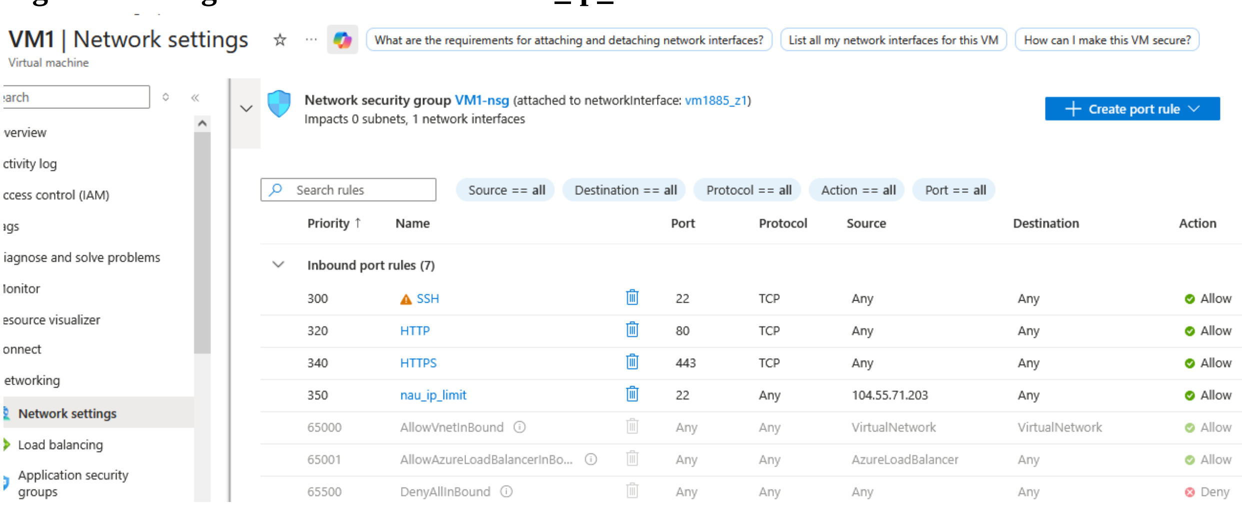
SSH attempt to VM1 (20.51.217.204) from PowerShell returns "Permission denied, please try again" followed by "Connection reset by 20.51.217.204 port 22." The nau_ip_limit NSG rule had a stale source IP that did not match the current public IP of the connecting machine.

Fig 2: VM1 Network Settings



VM1 Network Settings showing NIC vm1885_z1 on vnet-eastus-1 / snet-eastus-1, public IP 20.51.217.204, private IP 172.17.0.4, and NSG VM1-nsg. This is the entry point for all external SSH access.

Fig 3: VM1-nsg Inbound Rules with nau_ip_limit



VM1-nsg inbound rules showing SSH (300, Any), HTTP (320), HTTPS (340), nau_ip_limit (350, source 104.55.71.203, port 22, Allow), and DenyAllInbound (65500). Only the machine at 104.55.71.203 can SSH into VM1.

Fig 4: VM2 Overview Showing Unintended Public IP (Before Fix)

VM2 Overview at the time of the issue — Primary NIC public IP shows 172.174.155.67 and Virtual network/subnet shows vnet-eastus-2/snet-eastus-1. Both problems are visible here: VM2 is on the wrong VNet and has a public IP it should not have. VM2 was deleted and recreated to fix both issues.

Fig 5: VM2-nsg Inbound Rules with Internal_jump

Inbound port rules (4)

310	Internal_jump	22	Any	172.17.0.4	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBo...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

VM2-nsg inbound rules after recreation showing Internal_jump (priority 310, port 22, source 172.17.0.4, Allow). Only VM1's private IP can SSH into VM2. DenyAllInbound at 65500 blocks all other traffic.

Fig 6: ip addr Output on VM1 Confirming Private IP

```
craig@VM1:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 7c:1e:52:49:22:5e brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.4/24 metric 100 brd 172.17.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::7e1e:52ff:fe49:225e/64 scope link
       valid_lft forever preferred_lft forever
```

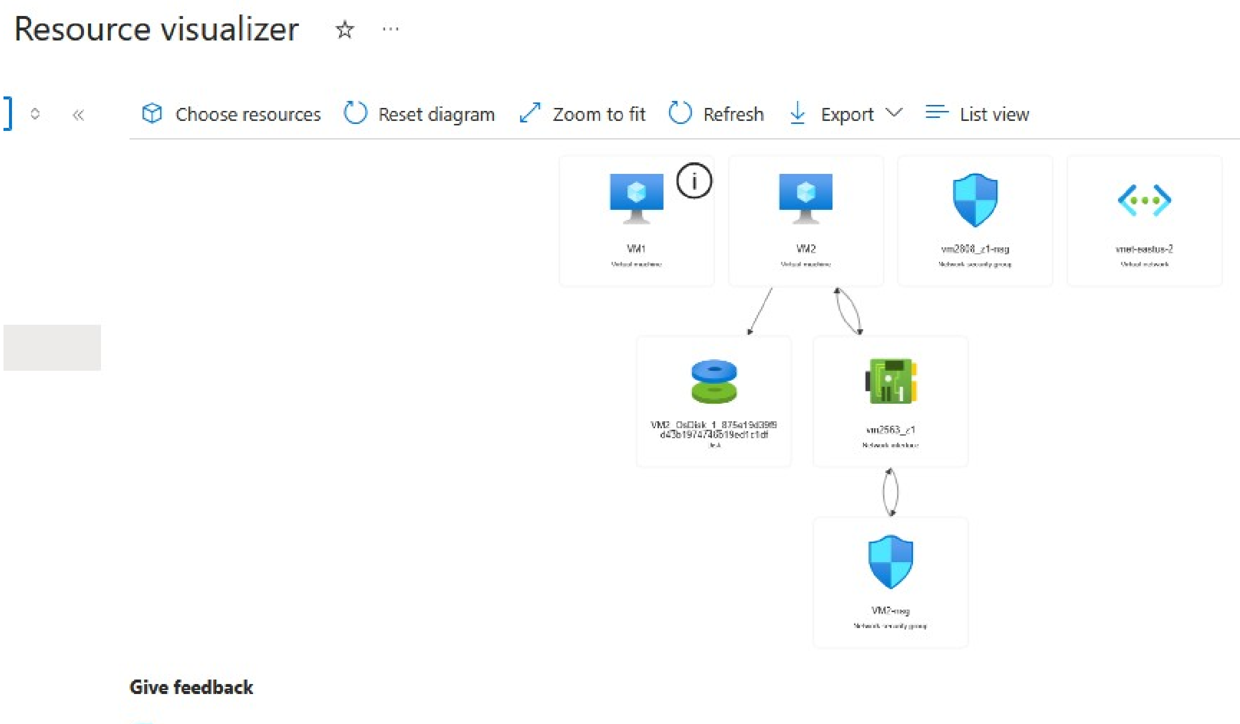
ip addr run inside VM1 shows inet 172.17.0.4/24 on the eth0 interface. This confirms VM1's private IP matches the source address entered in VM2's Internal_jump NSG rule.

Fig 7: Successful SSH from VM1 into VM2

```
craig@VM1:~$ ssh craig@172.17.1.4
The authenticity of host '172.17.1.4 (172.17.1.4)' can't be established.
ED25519 key fingerprint is SHA256:ejIQF/Y9xjh9sDCobTqAfLj7f7nfvoRHw7O2CODXhX4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.17.1.4' (ED25519) to the list of known hosts.
craig@172.17.1.4's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Feb 28 21:42:40 UTC 2026

  System load:  0.0               Processes:             125
  Usage of /:   5.6% of 28.02GB   Users logged in:       0
  Memory usage: 4%                IPv4 address for eth0: 172.17.1.4
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

craig@VM2:~$
```

SSH from VM1 to 172.17.1.4 succeeds. VM2's Ubuntu welcome banner confirms the connection and shows "IPv4 address for eth0: 172.17.1.4." The prompt changes to craig@VM2, confirming the jump-host path works: local machine → VM1 → VM2.

Fig 8: Resource Visualizer Showing Both VMs and Associated Resources



Azure Resource Visualizer showing VM1 and VM2 alongside VM2's OS disk, NIC (vm2563_z1), NSG (VM2-nsg), and the shared virtual network. Both machines are deployed and connected within the same resource group.

Challenges and Solutions

1. Stale NSG Source IP: The nau_ip_limit rule had an outdated IP. Resolved by checking the current public IP at whatismyipaddress.com and editing the rule.

2. Separate VNets: VMs on different VNets cannot communicate by private IP. Resolved by deleting VM2 and recreating it in vnet-eastus-1.

3. Unintended Public IP on VM2: Auto-assigned during creation. Resolved by explicitly setting Public IP to None when recreating VM2.

What I Learned

NSG Rules: Inbound rules with specific source IPs enforce access control at the Azure network layer. The DenyAllInbound rule at priority 65500 automatically blocks everything not explicitly allowed above it.

VNet Architecture: VMs must share a VNet to communicate via private IP. Resource groups are for organisation only and have no effect on connectivity.

Jump-Host Pattern: Isolating a private server (VM2) behind a public entry point (VM1) is a standard cloud security practice that prevents direct internet exposure of internal machines.

Troubleshooting: Reading Network Settings in the Azure portal to verify IPs and NSG rules, and using ip addr inside VMs to confirm actual interface addresses.

Conclusion

The lab successfully demonstrated NSG-based IP restriction for SSH on Azure. Three issues were identified and resolved: a stale source IP in the NSG rule, VM1 and VM2 being on separate virtual networks, and an unintended public IP on VM2. The final configuration restricts SSH on VM1 to a single authorized public IP and restricts SSH on VM2 to VM1's private IP only, implementing a secure jump-host architecture verified end-to-end with successful SSH sessions.